

Chapter 22

Symbols/Logic

1	2	3	4	5	6	7	8	9	10
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1) If $A \Omega S = A(A+S)$, then $4 \Omega 6 =$

- A) 12 B) 40 C) 6 D) 7 E) 24

2) If B is an integer > 1 and $A \odot B = \frac{A}{B\sqrt{A}}$ where $A \odot B$ is also a positive integer, what value of B would yield the largest value of $A \odot B$ for $64 \odot B$?

- A) 2 B) 4 C) 8 D) 64 E) 6

3) In the previous question, what is the value of $81 \odot 4$?

- A) 0 B) -44 C) 11 D) -2 E) 27

4) If $b \boxtimes t = 8x\left(\frac{b}{t}\right)$, what is the value of $3 \boxtimes 12$?

- A) 2 B) 202 C) 6 D) $\frac{1}{4}$ E) 180

5) If $A \otimes P = AP + P^2$, what is the value of $3 \otimes 12$?

- A) 2 B) 202 C) 6 D) $\frac{1}{4}$ E) 180

1	2	3	4	5	6	7	8	9	10
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6) Assuming $x \curvearrowright y = x\sqrt{y}$ and $x \curvearrowright \curvearrowright y = (x \curvearrowright y)^2$.

Which one of the following values of x would make the following equation true?

$$x \curvearrowright \curvearrowright 4 = 400$$

- A) 4 B) 12 C) 10 D) 100 E) 8

7) If A, B, C, X, Y, and Z are all positive integers where;

$Z \S A = 3\left(\frac{Z}{A}\right)$ and $Y \gg B = \frac{B}{Y}$ and $X \Delta C = \frac{X}{C}$. What is the value of

$$\frac{(12 \Delta 3)(5 \gg 10)(10 \S 5)}{(2 \gg 8)(14 \Delta 7)(3 \gg 18)}?$$

- A) 1 B) -1 C) $\frac{1}{2}$ D) $\frac{1}{4}$ E) 22

8) If $A \textcircled{R} B = A + 2B$. What is the value of $4 \textcircled{R} -3$?

- A) 0 B) 4 C) 11 D) -2 E) 1

9) If $p \times$ = the sum of the set of numbers that are prime numbers less than p where the **sum of the digits of each term is also prime**. What is the value of $250 \times - 230 \times$?

- A) -122 B) 241 C) 492 D) 233 E) 247

10) If $V \textcircled{\Delta} W = VW$ and $W \textcircled{+} V = W+V$, If $V = 2$ and $W = 3$ what is the value of $(V \textcircled{\Delta} W) \textcircled{+} W$?

- A) 5 B) 31 C) 9 D) 39 E) 54

1	2	3	4	5	6	7	8	9	10
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11) If $a \heartsuit b \spadesuit c = ab + ac$, what is $2 \heartsuit 3 \spadesuit 2$?

- A) 6 B) 10 C) 12 D) 7 E) 14

12) $L \heartsuit V = \frac{3V}{2L}$, If $8 \heartsuit V = \frac{9}{8}$ what is the value of V?

- A) 6 B) 10 C) 12 D) 7 E) 14

13) $P \odot D = P + D$ then $(4 \odot 0)^3 =$

- A) 64 B) 12 C) 16 D) 256 E) -512

14) If brown eyed cats never eat tuna then which of the following **must be false**?

- A) A brown-eyed cat did not eat tuna
 B) A brown-eyed cat ate tuna and chicken
 C) A brown-eyed cat ate chicken
 D) A red-eyed cat ate tuna
 E) A green-eyed cat ate chicken

15) If Brian rides his motorcycle to school, he will arrive early. Which of the following **cannot be true**?

- A) Brian rode the bus and was late for school
 B) Brian did not go to school
 C) Brian rode his motorcycle to school and was 1 minute late
 D) Brian got a ride to school from a friend and was early
 E) Brian rode his motorcycle to school and was 40 minutes early

Solutions Chapter 22

Symbols/Logic

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|------|-------|-------|
| 1) B | 6) C | 11) B |
| 2) E | 7) A | 12) A |
| 3) E | 8) D | 13) A |
| 4) A | 9) B | 14) B |
| 5) E | 10) C | 15) C |

1) When solving symbol problems, follow the key and substitute the values into the key. If $A \Omega S = A(A+S)$, the number appearing to the left of the symbol Ω is the A and the number appearing to the right of the symbol Ω is the S. First, add (A+S) as shown in the **key** then multiply the result by A; $A(A+S)$. For $4 \Omega 6$, the number to the left of the symbol Ω represents A; ($A = 4$) and the number to the right of the symbol represents S; ($S = 6$). Now substitute into the key $A(A+S)$, $4(4+6)$. To calculate, add 4 plus 6 equals 10 (PEMDAS), and by the distributive property, $4(10)$ equals 4 multiplied by 10 = 40.

40 Answer B Difficulty Level 3.5

2) Substitute the given information into the key $(A \odot B) = \frac{A}{B\sqrt[B]{A}}$; $64 \odot B = \frac{64}{B\sqrt[B]{64}}$, because the value before the symbol ($\odot$) is A = (64) and the value after the symbol ($\odot$) is B = (B). B represents the n^{th} root of 64. If B = 2, the denominator of the equation would equal the square root of 64 $\sqrt[2]{64}$ and if B = 3, the denominator of the equation would equal the cube root of 64 $\sqrt[3]{64} = 4$. Since B is a positive integer, B must equal 1, 2, 3, 4, 5, 6... but the only values of B that result in an integer are 2, 3, and 6. If B = 2, the square root of 64 is 8 and $\frac{64}{8} = 8$. (Note; normally, a 2 is not written in the radical to denote the square root. It is usually left blank). If B = 3, then the cube root of 64 is 4 and $\frac{64}{4} = 16$ and if B = 6, the 6th root of 64 is 2 and $\frac{64}{2} = 32$. Using 4 or 5 would give a non-integer value for B, and any value larger than 6 would also give a non-integer value.

6 Answer E Difficulty Level 10

3) When solving symbol problems, follow the key and substitute the values into the key; $\frac{A}{B\sqrt[4]{A}}$; For $(A\odot B)$; The number before the symbol $\odot$ represents $A(81)$, and the number after the symbol represents $B(4)$. Therefore, $81 \odot 4 = \frac{81}{\sqrt[4]{81}} = 81/3 = 27$. Recall the fourth root of $81 = 3$ because $3 \times 3 \times 3 \times 3 = 81$.

27 Answer E Difficulty Level 8

4) When solving symbol problems, follow the key and substitute the values into the key; $b \boxtimes t = 8 \times \left(\frac{b}{t}\right)$ The value before the symbol $\boxtimes$ is b , and the value after the symbol is t . For $3 \boxtimes 12$, $b = 3$ and $t = 12$. Plugging into the key $8 \times \left(\frac{b}{t}\right)$, $8 \times \left(\frac{3}{12}\right) = \frac{24}{12} = 2$.

2 Answer A Difficulty Level 3.5

5) When solving symbol problems, follow the key and substitute the values into the key. $A \otimes P = AP + P^2$ where A is the value before the key and P is the value after the key. Find the value of $3 \otimes 12$ where $A = 3$ and $P = 12$ by plugging into the key $3(12) + (12)^2 = 36 + 144 = 180$

180 Answer E Difficulty Level 3

6) When solving symbol problems, follow the key and substitute the values into the key. To solve $x \wr \wr \wr 4 = 400$ for x for the first part of this question, use the second key shown; $x \wr \wr y = (x \wr y)^2$ where $x = x$ and $y = 4$. Plugging these values into the key obtains $(x \wr 4)^2$. Now use the first given key; $x \wr y = x\sqrt{y}$ and substitute. $(x\sqrt{4})^2 = 400$. Solve for x by first simplifying the left side of the equation, $(2x)^2 = 4x^2$; So $4x^2 = 400$. Divide each side of the equation by 4 to get $x^2 = 100$. Take the square root of both sides of the equation; $x = 10$ or -10 . When solving an equation by taking an even root of both sides of an equation, produces two answers; one positive and one negative. Since -10 is not one of the answer choices, $x = 10$.

10 Answer C Difficulty Level 8.5

7) When solving symbol problems, follow the key and substitute the values into the key. This question has three different keys. Use the correct key corresponding to its matching symbol. Break up the fraction piece by piece; $(12 \blacktriangle 3)$ where $X = 12$ and $C = 3$, $(12 \blacktriangle 3) = \frac{12}{3} = 4$, $(5 \gg 10)$ where $Y = 5$ and $B = 10$, $(5 \gg 10) = \frac{10}{5} = 2$, $(10 \S 5)$ where $Z = 10$ and $A = 5$, $(10 \S 5) = 3\left(\frac{10}{5}\right) = 6$, $(2 \gg 8)$ where $Y = 2$ and $B = 8$, $(2 \gg 8) = \frac{8}{2} = 4$, $(14 \blacktriangle 7)$ where $X = 14$ and $C = 7$, $(14 \blacktriangle 7) = \frac{14}{7} = 2$, and $(3 \gg 18)$ where $Y = 3$ and $B = 18$, $(3 \gg 18) = \frac{18}{3} = 6$. Finally plugging in; $\frac{(4)(2)(6)}{(4)(2)(6)} = \frac{48}{48} = 1$. Notice the **same numbers** appear in the numerator and denominator and could have been cross canceled to reduce.

1 Answer A Difficulty Level 7

8) Substitute the given into the key. The value before the key $A (\textcircled{R}) = 4$ and the value or number after the key $(\textcircled{R}) B = -3$. Since $A \textcircled{R} B = A + 2B$, substitute into the equation, $4 + 2(-3) = 4 - 6 = -2$

-2 Answer D Difficulty Level 2.5

9) To answer this question correctly, exercise multiple topics. First, when subtracting all the prime numbers in the second set 230 X from all the prime numbers in the first set 250 X , the remaining prime numbers are between 230 and 250. Notice, each number in the first set that is prime with the sum of their digits also prime is canceled by the same numbers in the second set when they are subtracted. (Remember prime numbers are numbers greater than one and are only divisible by one and themselves).

$\{2 + 3 + 5 + 7 + 11 + \dots 223, 227, 229, \textbf{233, 239, 241}\} -$

$\{2 + 3 + 5 + 7 + 11 + \dots 223, 227, 229\}$ leaving **233, 239, and 241**.

233 and 239 are both prime numbers but the sum of their digits are both not prime.

233; $2 + 3 + 3 = 8$ is not prime and

239; $2 + 3 + 9 = 14$ is also not prime.

That leaves 241.

How to check if a number is prime? All prime numbers larger than 3 can be expressed as either $6n + 1$ or $6n - 1$, where n is a positive integer.

Check 231;

$$6n + 1 = 231, 6n = 230, n = 38.33 \text{ decimal, or}$$

$$6n - 1 = 231, 6n = 232, n = 38.67 \text{ decimal,}$$

thus, 231 must not be prime (composite).

Now check 233;

$$6n + 1 = 233, 6n = 232, n = 38.67 \text{ decimal, or}$$

$$6n - 1 = 233, 6n = 234, n = 39 \text{ with no remainder.}$$

Therefore, 233 is prime.

241 Answer B Difficulty Level 9.5

10) When solving symbol problems, follow the key and substitute the values into the key. This question has two different keys. $V \heartsuit W = VW$ and $W \odot V = W + V$. Use the correct key corresponding to its matching symbol. Next, use the rules of PEMDAS by starting with P parenthesis $(V \heartsuit W) \odot W$. Since $V = 2$ and $W = 3$ then $V \heartsuit W = VW = 2(3)=6$. Finally, $6 \odot W$ or $6 \odot 3 = 6+3 = 9$.

9 Answer C Difficulty Level 4.5

11) When solving symbol problems, follow the key and substitute the values into the key. $a \heartsuit b \spadesuit c = ab + ac$, where a is the value before the $\heartsuit$, b is the value after the $\heartsuit$, and c is the value at the end after the spade $\spadesuit$. So, $a = 2, b = 3$, and $c = 2$. Finally, $2 \heartsuit 3 \spadesuit 2 = 2(3) + 2(2) = 6 + 4 = 10$

10 Answer B Difficulty Level 4

12) When solving symbol problems, follow the key and substitute the values into the key. $L \heartsuit V = \frac{3V}{2L}$. Where L is the value or number(8) in this case before the symbol $\heartsuit$ and V is the value after the symbol $\heartsuit$, in this case, it is also V . Therefore $L \heartsuit V = \frac{3V}{2(8)} = \frac{9}{8}$ or $\frac{3V}{16} = \frac{9}{8}$. Now, cross multiply to get $24V = 144$ and $V = 6$.
Remember, cross multiply only when there is an equal sign between two fractions.

6 Answer A Difficulty Level 7.5

13) When solving symbol problems, follow the key and substitute the values into the key. For $P \oplus D$, the value before the symbol ($\oplus$) is P, and the value after the symbol ($\oplus$) is D. For $4 \oplus 0$, $P = 4$ and $D = 0$. Plugging into the key;
 $P \oplus D = P + D$, $4 \oplus 0 = 4 + 0 = 4$. Therefore, $(4 \oplus 0)^3 = (4 + 0)^3 = (4)^3 = 64$.

64 Answer A Difficulty Level 3

14) This question can be correctly answered by eliminating answers that do not make sense. In this question, the correct answer choice is the false one. The statement only refers to cats with brown eyes, making it impossible to make inferences about cats with any other eye color. Eliminate answer choices D and E because no conclusion can be made about cats with red or green eyes.

Answer choice A is incorrect because it must be true; brown-eyed cats never eat tuna. Answer choice C is incorrect because brown-eyed cats never eat tuna. Nothing was stated saying these brown-eyed cats could not eat chicken. That leaves the correct answer choice B because brown-eyed cats never eat tuna and a the brown-eyed cat ate tuna and chicken, which must be false.

A brown-eyed cat ate tuna and chicken

Answer B Difficulty Level 6

15) In this question, the correct answer is the answer choice that cannot be true.

The statement only refers to what happens when Brian rides his motorcycle to School. If Brian does not ride his motorcycle to School, no conclusions can be made. No conclusion can be made for answer choice A because Brian rode the bus to School. Answer choice B makes no sense because we have no idea if Brian went to School. No conclusion can be made for answer choice D because Brian got a ride from a friend and did not ride his motorcycle to School. Answer choice E is not correct because it can be true from the original statement. Since Brian rode his motorcycle to School he is supposed to arrive early. This leaves the correct answer choice, C because Brian did ride his motorcycle to School, but Brian arrived late. This contradicts the original statement and cannot be true.

Brian rode his motorcycle to School and was 1 minute late

Answer C Difficulty Level 6